Biennial Progress Report on

Operational Probabilistic Coordinated Security Assessment and Risk Management

December 2025

entsoe

Foreword

ENTSO-E, the European Network of Transmission System Operators for Electricity, is the association of the European transmission system operators (TSOs). The **40 member TSOs, representing 36 countries,** are responsible for the secure and coordinated operation of Europe's electricity system, the **largest interconnected electrical grid in the world.**

Before ENTSO-E was established in 2009, there was a long history of cooperation among European transmission operators, dating back to the creation of the electrical synchronous areas and interconnections which were established in the 1950s.

In its present form, ENTSO-E was founded to fulfil the common mission of the European TSO community: to power our society. At its core, European consumers rely upon a secure and efficient electricity system. Our electricity transmission grid, and its secure operation, is the backbone of the power system, thereby supporting the vitality of our society. ENTSO-E was created **to ensure the efficiency and security of the pan-European interconnected power system** across all time frames within the internal energy market and its extension to the interconnected countries.

ENTSO-E is working to secure a carbon-neutral future. The transition is a shared political objective through the continent and necessitates a much more electrified economy where sustainable, efficient and secure electricity becomes even more important. **Our Vision: "a power system for a carbon-neutral Europe"*** shows that this is within our reach, but additional work is necessary to make it a reality.

In its Strategic Roadmap presented in 2024, ENTSO-E has organised its activities around two interlinked pillars, reflecting this dual role:

- "Prepare for the future" to organise a power system for a carbon-neutral Europe; and
- "Manage the present" to ensure a secure and efficient power system for Europe.

ENTSO-E is ready to meet the ambitions of Net Zero, the challenges of today and those of the future for the benefit of consumers, by working together with all stakeholders and policymakers.

* https://vision.entsoe.eu/

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Executive Summary

Consistent with the legal mandate, the *2025 Biennial operational probabilistic coordinated security assessment and risk management Progress Report* (hereafter the *2025 PRA Report*) provides a view of all TSOs' progress towards an operational probabilistic coordinated security assessment and risk management. This is pursuant to the methodology for coordinating operational security analysis (CSAM [3]), specifically Articles 44(1) and 44(2).

Following the Agency for the Cooperation of Energy Regulators' (ACER) decision establishing the CSAM in June 2019, TSOs have set up sequential governance structures within ENTSO-E's steering group operational framework and under the guidance of the System Operations Committee to prepare for and plan the move towards probabilistic risk assessment (PRA).

The 2025 PRA Report is the third public report detailing progress, expected challenges and the next steps towards developing a PRA methodology by the end of 2027. The ENTSO-E Working Group Probabilistic Risk Assessment (WG PRA) has been established to develop a robust probabilistic risk assessment approach. Since the last biennial report, WG PRA has been investigating and testing probability computation methods while developing a proof-of-concept (POC) for the impact assessment. In parallel, WG PRA continues to improve the PRA data collection process to ensure all necessary data are available and reliable.

Progress

The first biennial report was published in 2021, and since then, WG PRA has made considerable progress in developing the PRA. The first version of the Grid Disturbance Profile has been developed, enabling harmonised reporting of grid disturbances while remaining compatible with the common information model (CIM).

The document is currently being updated. Furthermore, the TSOs have successfully implemented a new IT infrastructure for the PRA data collection process, providing PRA data on this platform since 2023. With respect to the PRA methodology development, WG PRA identified several methodologies for probability computations, successfully tested the so-called VAFFEL (varsel før feil; English: warning before failure) concept[1] and started developing the POC for the impact assessment.

1 The VAFFEL concept was developed by Statnett and introduced in the 2023 PRA Biennial Progress Report. Details on the concept can be found in [1] and [5].

Challenges

Changes in TSOs' operational processes – particularly at the pan-European level – are accompanied by challenges and hurdles. Each biennial report presents TSO responses to survey questions, enabling the identification of the key challenges that TSOs foresee in developing and implementing a PRA methodology. These challenges include:

- changes to current TSOs' internal, regional and pan-European processes, requiring a move from a well-proven and comprehensible deterministic approach to a more complex probabilistic one;
- the need for additional investment (labour and IT) to address a move towards PRA;
- managing an increased volume of data while simultaneously ensuring that the data is secure, of high quality and reliable to build the PRA methodology;
- ensuring the development of a methodology consistent with the legal mandate, striking a balance between complexity, practicality, network security and socioeconomic benefits.

These challenges will be considered while developing the PRA methodology and will be further discussed with the TSOs.

Next Steps

WG PRA continuously engages with TSOs to ensure the PRA methodology addresses the identified challenges and prepares TSOs for its future implementation in their processes.

Over the following years until the delivery of the PRA methodology, WG PRA will continue developing the impact assessment approach and testing the probability calculations that underpin PRA. Workshops will be organised with TSOs to discuss the first results of the POC, and the methodology will be reviewed based on the feedback received.

WG PRA will continue to support TSOs in the data collection process, ensure quality management by validating the data on the platform and update the PRA dataset to ensure that relevant data for the risk assessment are collected.

Additionally, WG PRA will ensure complete alignment with ACER in a dedicated workshop to be planned in the first semester of 2026 to present the status of work and discuss the next steps.

1 Introduction

Historically and currently in Europe, power system operational security management has relied on the 'N-1' criterion[2] as the criterion governing security assessment. Accordingly, the power system is always able to withstand an unexpected failure or outage of a system component while accommodating the new operational situation without violating existing security limits.

PRA is a complementary operational security management approach that allows TSOs to consider the probability and subsequent impact of power system failures when establishing their security limits, reflecting an expansion of existing methods security limits. This is an expansion to existing methods which are based on the 'N-1' criterion. The N-1 criterion assumes that all disturbances and failures in the N-1 contingency list are equally likely and that none of these contingencies should lead to operational limit violations (regardless of their impact).

Establishing the PRA approach entails quantifying the system's expected performance while accounting for uncertainties in its operational conditions (e.g., weather conditions over a specified period).

The legal obligation for TSOs to develop a PRA methodology is defined in CSAM Article 44:

CSAM Article 44 – 'Towards probabilistic risk assessment'

Art. 44.1 – By 31 December 2027, all TSOs shall jointly develop the methodology on common probabilistic risk assessment taking full account of the requirements of Article 75(1)(b) and Article 75(5) of the SO Regulation, and shall propose it as an amendment of this methodology in accordance with Article 7(4) of the SO Regulation. After its approval in accordance with Article 7 of the SO Regulation, the methodology on common probabilistic risk assessment shall form an annex to this methodology.

The SO Regulation Article 75(1)(b) requires that the CSAM shall at least cover the principles for common risk assessment for the contingencies referred to in Article 33 and SO Regulation Article 75(5) specifies that those principles shall include – among others – the evaluation of the probability and impact of exceptional contingencies.

SO Regulation Article 33 determines how TSOs shall establish their contingency list based on a classification of each contingency as ordinary, exceptional or out-of-range.

For this purpose, ENTSO-E – with the support of TSOs – has created WG PRA, whose main goal is to develop the PRA methodology by 2027. This work was initiated in 2019, and the achievements made are presented in the dedicated 2021 and 2023 PRA Biennial Progress Reports.

This report aims to present the accomplishments achieved by WG PRA – with the support of TSOs – towards the development of the PRA methodology in 2024 and 2025.

2 Throughout this report, the 'N-1' criterion will be considered pursuant to System Operation Guideline definitions, that is, 'N-1' refers to the N-state minus 1 contingency. Each contingency can consist of one element (ordinary contingency) or of several elements (exceptional contingency).

1.1 PRA Progress Report 2023

The second biennial report was issued in December 2023. It presented:

- The ENTSO-E WG PRA, which was established in April 2021, presenting the objectives, structure and focus area of its three workstreams.
- The survey conducted in 2023 to assess TSOs' preparedness for PRA implementation. The survey highlighted ongoing challenges, including changes to current processes, data volume and quality and the need for significant investment. The progress achieved compared to 2021 survey was presented.
- The risk was defined as the product of probability and impact. The PRA methodology aimed to enhance grid operation efficiency by providing detailed risk information. Historical weather and failure data, along with forecasts, were presented for use in improving CSA processes. The VAFFEL concept was presented, which aims to calculate failure probabilities for overhead lines based on weather forecasts. The impact assessment involved evaluating loss of load and energy not supplied (ENS) during contingencies, requiring updates to current security analyses.
- The improvements regarding the PRA data collection. The first version of the Grid Disturbance Profile was presented, along with the extension of the existing ENTREC platform to upload and validate PRA data.

The reader can refer to the 2021 and 2023 PRA Biennial Progress Reports for additional information on these topics.

1.2 2025 Biennial PRA Report outline

The 2025 Biennial PRA Report is divided into four chapters:

Chapter 1 introduces the report, briefly explains what PRA is and outlines its expected benefits. It also describes the outcome of the 2023 Biennial Report.

Chapter 2 presents the organisation of WG PRA established at ENTSO-E, which is responsible for developing the PRA methodology by 2027. The focus areas and achievements of each workstream over the past two years are presented.

Chapter 3 describes the achievements made in these years. First, the improvements made by TSOs based on a survey similar to the 2023 survey are summarised. Subsequently, the progress achieved by WG PRA in developing the PRA methodology – impact and probability computations – and in data collection is detailed.

Chapter 4 presents other PRA activities initiated by some TSOs.

Chapter 5 presents the roadmap for moving towards probabilistic risk assessment methodology by 2027 and clarifies the target steps for the next Biennial Report 2027.

2 Working Group Probabilistic Risk Assessment

ENTSO-E has established WG PRA as a dedicated working group to fulfil the long-term mandate of the PRA development and ensure continuity and knowledge retention for such a complex topic.

The terms of reference for WG PRA were approved in April 2021 and revised in November 2023, with the main objective of supporting TSOs in fulfilling their PRA-related mandates. The size of the working group has increased since 2021, now encompassing 20 TSOs and two regional coordination centres (RCCs).

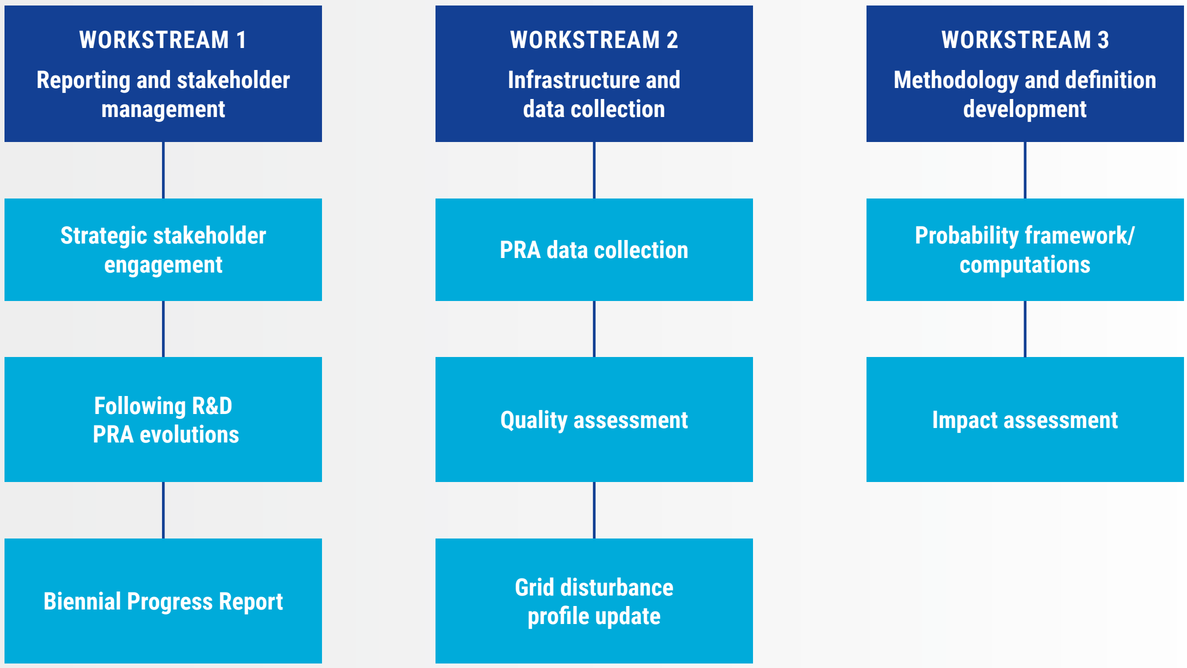


Figure 1: WG PRA's three workstreams with their respective focus areas

The objectives of WG PRA are to:

- develop the probabilistic risk assessment methodology and proactively provide expertise and support the TSOs on its implementation and interpretation;
- develop and set up – together with RCCs – the infrastructure required to collect and process the data for probabilistic risk assessment;
- manage the changes and necessary amendments on CSAM entailed by the probabilistic risk assessment methodology and assess whether it is necessary to amend the operational network codes and guidelines (i.e., SO regulation);
- deal with all other relevant network code regulatory issues related to probabilistic risk assessment, including the implications and impact on system operation from/to the connection codes and market codes;
- proactively follow the development of new regulations and monitor developments in the field of probabilistic risk assessment;
- develop and publish (on ENTSO-E's website) a report on the progress achieved in Europe on the operational probabilistic coordinated security assessment and risk management;
- manage and facilitate a constructive dialogue between ENTSO-E, stakeholders and regulatory authorities.

WG PRA is divided into three workstreams, as shown below. As mentioned in the previous report, the workstreams' focus areas are revised in line with the year's objectives.

Workstream 1: Reporting and stakeholder management

Workstream 1 is responsible for the biennial report, stakeholder engagement and proactive engagement with industry evolutions. During 2024 – 2025, the focus has been on mapping all TSOs' activities in the field of probabilistic assessment to identify potential synergies with WG PRA.

Workstream 2: Infrastructure and data collection

Workstream 2 is responsible for continuous ENTSO-E infrastructure development, improvement and monitoring to support data collection in support of the move towards PRA. In 2025, the working group updated the grid disturbance definitions for the power system above 100 kV [2] to refine and improve clarity. The new version will be published on ENTSO-E's website in 2026. Moreover, Workstream 2 has supported TSOs in the annual collection of the PRA grid disturbance data.

Workstream 3: Methodology and definition development

Workstream 3 has been further divided into two substreams. Workstream 3A has been focusing on performing and/or evaluating probability calculations with various available methods. An important task was to work with Workstream 2 to determine which collected data are required for the calculations.

Workstream 3B determined the operational data needed for PRA and added them to the collection process under the responsibility of Workstream 2. Furthermore, Workstream 3B has been exploring the impact assessment and the general flowchart of the PRA methodology. The results of both workstreams will later be combined into a risk assessment methodology.

3 Progress on Operational Probabilistic Coordinated Security Assessment and Risk Management

This chapter provides a more detailed view of the progress achieved in operational probabilistic coordinated security assessment and risk management, highlighting developments, reviewing potential future hurdles and obstacles and assessing the necessary steps and precautions for developing the methodology for common probabilistic risk assessment by 2027.

3.1 Progress achieved compared to 2023 levels

As previously mentioned, to assess all TSOs' preparedness for PRA, WG PRA conducted a survey covering expected benefits and challenges and hurdles in implementing PRA at the pan-European level (with a particular focus on data collection practices), as well as questions about PRA's progress to date and prospects. The 2025 survey's questions were extended from similar surveys conducted in 2023, 2021 and 2019.

A comparison between 2025 and the previous surveys identified that several TSOs achieved progress in different areas related to probabilistic risk assessment, which can be summarised as follows:[3]

- one TSO progressed in collecting the data about weather conditions at the time of the contingency;
- six TSOs have improved their data collection processes and are now fully ready to collect information about fault causes according to the ENTSO-E Grid disturbance definitions for the power system above 100 kV;
- the same six TSOs are also ready to collect information associated with grid disturbances consistent with the definition according to the ENTSO-E grid disturbance definitions for the power system above 100 kV;
- six other TSOs have improved their data collection processes and are now fully ready to collect information about the faults, disturbances and outages according to ENTSO-E grid disturbance definitions for the power system above 100 kV.

3 21 TSOs provided answers to the 2023 survey.

The 2025 survey also introduced new questions, leading to the following findings among participating TSOs

- TSOs were asked about any other data related to the occurrence of contingencies on the grid. Some of them collect information from protection devices or concerning weather conditions, or conduct detailed incident analyses in specific cases.
- TSOs were asked which system security criteria they would include in their impact assessment tool, and the most commonly used criteria were thermal and voltage limits, N-1 security, ENS, system stability and risk of cascading failures. While some TSOs would apply advanced tools and dynamic assessments, others would face challenges due to limited involvement, unclear methodologies or a lack of familiarity with impact assessment tools. Weather conditions and the performance of the protection system would also be considered in some cases.
- TSOs were asked how they would define the thresholds for acceptable and unacceptable risks, and most TSOs have not yet formally defined those thresholds, often relying instead on the N-1 criterion, operational security limits or qualitative assessments based on potential consequences such as ENS, system instability or equipment damage.
- Finally, TSOs were asked about the key sources of uncertainty in their day-ahead and intraday congestion forecasts. Most of them identify renewable generation forecasts, weather conditions, load variations and market behaviour as the primary sources of uncertainty in day-ahead and intraday security assessments.

According to the survey results, TSOs believe that the PRA methodology can improve risk assessment, optimise grid utilisation, complement deterministic methods in the future and reduce costs.

However, as identified in the previous report, the main barriers to implementing a PRA approach in power system operations revolve around the lack of high-quality historical data, computational and methodological complexity and the difficulty of integrating PRA into current operational processes. Concerns also include the acceptance of a probabilistic mindset by operators, authorities and the public, especially when dealing with low-probability, high-impact events. Additional challenges involve defining acceptable risk thresholds, interpreting probabilistic results under time pressure and accurately simulating cascading failures.

TSOs were asked about potential obstacles to implementing additional IT infrastructure (software, databases, etc.), its integration with the existing IT infrastructure and the adaptation of the existing IT infrastructure to new requirements. Most of them face significant obstacles to implementing new IT infrastructure for disturbance reporting, mainly due to limited resources, competing priorities and complex procurement or integration processes. They were also asked about potential obstacles to the extraction, collection and processing of data on faults, disturbances and outages (including a lack of data and low data quality). They highlighted limited resources, fragmented data systems, inconsistent formats and a lack of internal acceptance, making it difficult to collect, standardise and maintain high-quality fault and disturbance data for PRA purposes.

To mitigate these identified risks, there is a clear need for collaboration among the TSOs to ensure that:

- The PRA methodology strikes a balance between complexity, practicality, network security and socioeconomic benefits. The development of the POC will help clarify the added value of the PRA methodology, and the results will be discussed to adapt the methodology based on the feedback received. In its first implementation phase, PRA will be used in parallel with the 'N-1' criterion to provide operators with more insights into the network. It is also intended that the methodology's first implementation would help TSOs assess the risk level at which they are operating their grids at present, helping them align on an acceptable risk level.
- A large and reliable PRA database is available for the TSOs. The PRA data collection began at the start of the project to support the TSOs in collecting and improving the quality of their data. The PRA dataset will be continuously reviewed so TSOs can focus their collection process on the data required for the PRA methodology. By starting the process early, it is intended that the data will become increasingly reliable over the years, as will the PRA methodology.
- A balance is found between an efficient PRA methodology and a reasonable investment for all TSOs. The PRA implementation timeline needs to align with TSOs' ability to adapt their operational processes and related IT infrastructure.

3.2 Progress of WG PRA on the PRA methodology

The development of the PRA methodology is pursuant to CSAM Article 44, especially Article 44.4, which states that: *"… all TSOs shall jointly develop the methodology on common probabilistic risk assessment taking full account of the requirements of* ***Article 75(1)(b)*** *and* ***Article 75(5)*** *of the SO Regulation [...]."*

The requirements of the SO Regulation [4] are presented below.

SO Regulation Article 75 – 'Methodology for coordinating operational security analysis'

Art. 75.1. By 12 monthsA after entry into force of this Regulation, all TSOs shall jointly develop a proposal for a methodology for coordinating operational security analysis. That methodology shall aim at the standardisation of operational security analysis at least per synchronous area and shall include at least:

(b) principles for common risk assessment, covering at least, for the contingencies referred to in Article 33:

(i) associated probability;

(ii) transitory admissible overloads; and

(iii) impact of contingencies.

Art. 75.5. The principles for common risk assessment referred to in point (b) of paragraph 1 shall set out criteria for the assessment of interconnected system security. Those criteria shall be established with reference to a ***harmonised level of maximum accepted risk*** *between the different TSO's security analysis. Those principles shall refer to:*

(a) the consistency in the definition of exceptional contingencies;

(b) the evaluation of the probability and impact of exceptional contingencies; and

(c) the consideration of exceptional contingencies in a TSO's contingency list when their probability exceeds a common threshold.

The core of the PRA methodology lies in the term risk, which can be calculated for each contingency or summed at the regional level for any system state. Risk is the product of **the probability of an event occurring** and its **impact**.

It is thus expected that the PRA methodology would allow mapping all contingencies by their related **risk** and comparing them against a harmonised level of **acceptable risk**, as shown in Figure 2. All contingencies located in the 'unacceptable risk' area would constitute a PRA risk screening list that would need to be addressed through remedial actions. In the future, the current deterministic N-1 approach to operational security could be substituted by a PRA assessment to determine the PRA risk screening list to be covered.

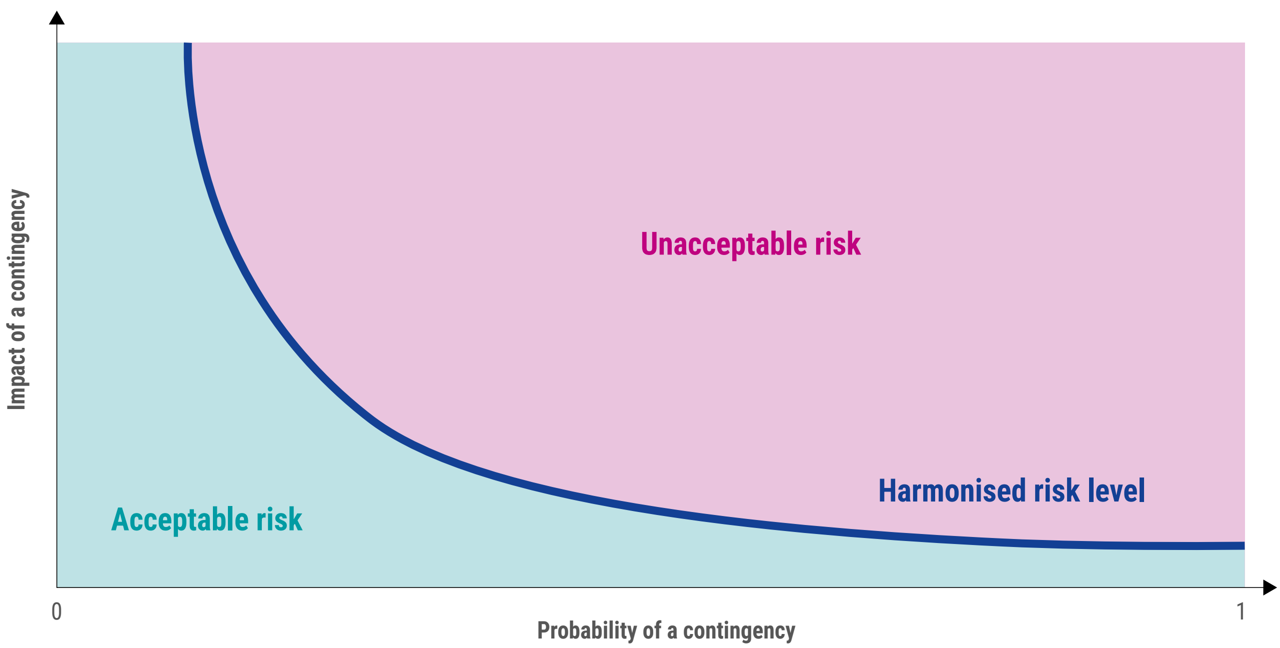


Figure 2: Risk assessment

3.2.1 PRA expected outcome

The purpose of the PRA methodology is to ensure that all TSOs handle contingencies in a harmonised, more realistic way. Indeed, by accounting for failure probabilities (dependent on weather, maintenance schedules, fires, etc.) and computing the impact to assess the risk of each contingency, PRA would provide TSOs with a more comprehensive and realistic view of what could happen in real-time operations.

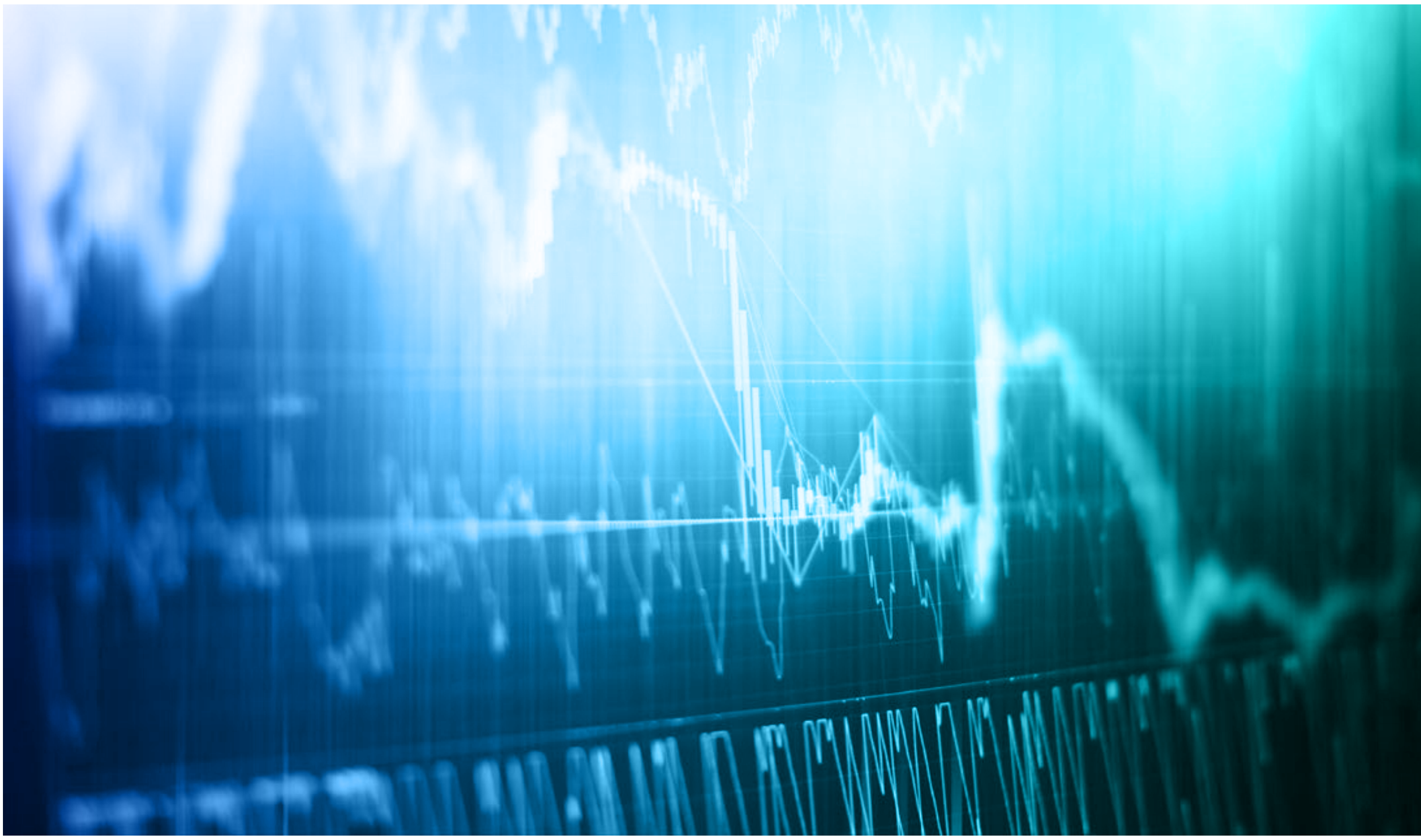
The N-1 deterministic approach is simple and easier to compute, but does not provide the TSO with a view of the risk associated with grid security and how remedial actions put in place mitigate this risk. The N-1 approach does not allow for a direct quantitative comparison between two N-1-secure system states, whereas, with the PRA methodology, if the risk is non-zero, it provides a quantitative indicator for comparing two secure states. The risk also indicates to operators whether the secure state is close to the system limits.

It is expected that the PRA methodology would allow the current power system to operate more efficiently and closer to its capacity by better understanding the risk. Nevertheless, in a highly meshed system, an unsolved N-1 may nearly always lead to a cascade that impacts most of the system. In this case, regardless of the probability of the incident, the impact will never be acceptable, and PRA will not add more value than the deterministic approach.

Of course, a POC should be developed to conclude on the benefits of the PRA methodology. Depending on the system's characteristics, the PRA risk screening list established using the PRA methodology might yield results similar to those obtained by applying the N-1 criterion (i.e., the unacceptable contingencies for which remedial actions are required are the same in both methods). Thus, it is important to assess how the theoretical concepts of the PRA methodology affect the establishment of the contingency list for several scenarios of the pan-European system in practice. Comparing the contingency lists as determined today with the PRA risk-screened list would allow the TSOs to evaluate whether (and under what conditions) the current rules for establishing the contingency list are suitable.

For this purpose, WG PRA aims to develop the POC of the PRA methodology. For this first step, WG PRA intends to focus on day-ahead and intraday security assessments, which enable focusing on forecasts of input variables that reflect the best-estimated values for those variables. The long-term timeframe would require the consideration of uncertainties.

The main challenge for the development of the POC is calculating probability and impact, which will be described in more detail in the following.



3.2.2 Probability calculations

Many factors influence the probability of failure of a grid component, as presented in the CSAM Article 8. The computations for the probability of failures should take at least the following into consideration:

- history of incidents impacting an element;
- weather and environmental conditions;
- specific geographical location;
- technical conditions;
- time of year (summer/winter);
- ongoing maintenance work (work in stations increases the probability of faults, especially on control equipment);
- lead time (e. g., week-ahead, day-ahead).

To compute the probabilities, each TSO should rely on a detailed database of their grid's failure history. WG PRA collects the relevant information to support the TSOs (see Chapter 3.3).

WG PRA is currently investigating the methodologies available to compute the probabilities. Based on the selected methodologies, WG PRA will consistently review the PRA dataset collected from the TSOs to ensure that all relevant data are collected for the failure computation.

Such computations are challenging because they rely on high-quality input. Additionally, the mathematical developments need to capture the specificities of each failure cause and will most likely be based on assumptions that need to be carefully addressed to ensure the output of the computations is reliable for use in an operational security assessment.

The main task of Workstream 3A during the period since the last biennial report was to identify, evaluate and – if possible – test available methods for the probability calculation. Some of these methods focus only on probability computations, while others propose a PRA method, as presented in Chapter 4.

The main effort was focused on the VAFFEL concept [1] and [5] since the source code is available to WG PRA. The concept focuses on computing failure probability due to wind and lightning. MAVIR and TenneT NL tested the method with their own data and verified that:

- The concept can be implemented in a TSO's own environment and – given that proper data is available – used to evaluate outage occurrence. MAVIR's investigations show that two events (one due to wind and the other due to lightning) could have been forecasted using the method.
- The concept can be easily extended to consider other factors if a proper mathematical formulation is developed. MAVIR included the effect of icing in the computations, although outages cannot be accounted for using the icing engine due to the lack of proper data. This underscores the importance of having a large, reliable dataset on which to perform the probability computations.

Other evaluated methodologies require code development to be tested. In the near future, WG PRA aims to test additional methodologies while ensuring that the necessary data for probability computations are collected.

In the upcoming period, the task of this subgroup is to test additional methods – if possible – and collaborate with Workstream 2, focusing on data collection to ensure that the necessary data for probability computations are collected.

3.2.3 Impact assessment

An impact assessment is necessary to compute the risk associated with a contingency. The impact can be defined in various ways, based on indicators such as the simulated ENS, the value of lost load (VoLL) and component overloads.

The impact assessment within the scope of a PRA differs from the current N-1 deterministic approach, which primarily focuses on overloads following a contingency. The PRA approach would assess the end-state impact of a contingency by asking: What will the final system state be after the contingency if no remedial actions are taken? What happens if operators allow the overload after a contingency? Will it create further constraint violations? The answers to these questions will most likely vary from one contingency to another.

The previous biennial report presented the first flowchart of the impact assessment, divided into two steps:

- **Step 1:** Determining the final state of the power system after a contingency (without any manual intervention).
- **Step 2:** Computing the indicators on the final state.

Figure 3 presents a flowchart for such an assessment, aiming to compute the impact indicators. WG PRA will investigate which impact indicator(s) are most suitable to assess the impact of a contingency in the scope of a probabilistic assessment. The SPS step refers to the activation of system protection schemes (SPSs).

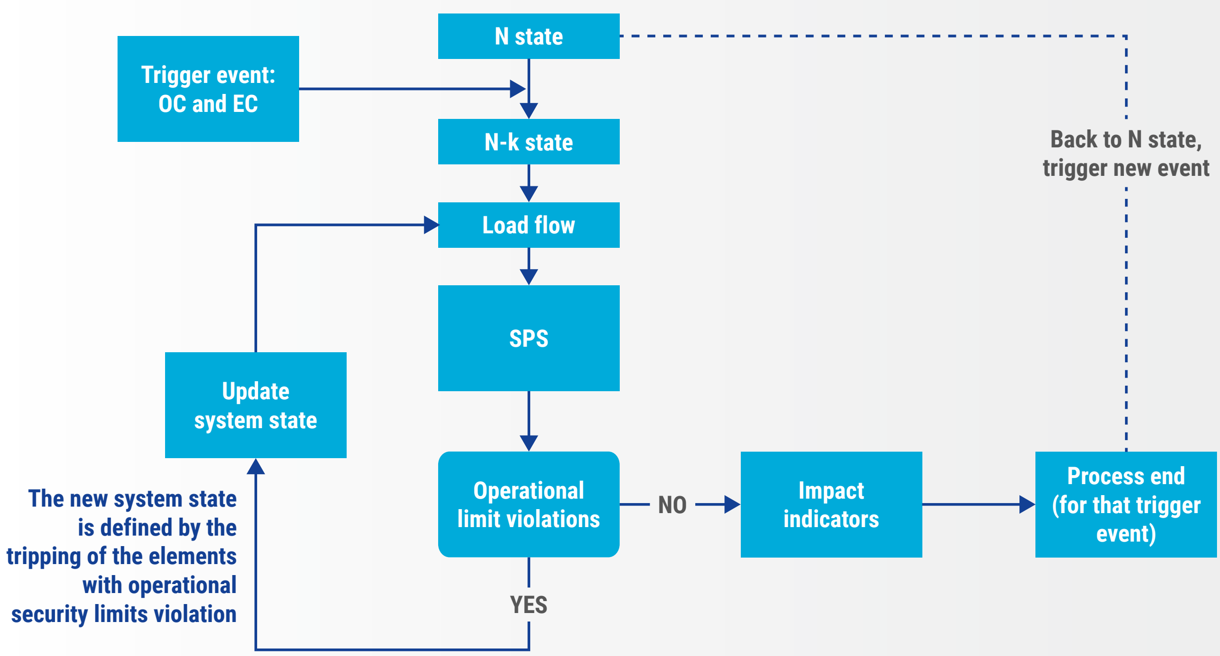


Figure 3: High-level impact assessment flowchart

Based on this flowchart, Workstream 3B is developing a POC of the procedure for the first step, which assesses system state and operational security limit violations.

Model description

The model is developed in Power Factory, and the Python API is used to run the simulations. The model aims to simulate the flowchart from Figure 3 with the following considerations:

- The grid models used for the simulations are DACF CGMs provided by TSCNET.
- SPSs are not modelled yet.
- The simulations stop when the power flow results show no further operational violations, or when the power flow calculation does not converge, in which case the impact is set to be unacceptable. Ultimately, the results show the ENS and all other operational variables.
- Both ordinary and exceptional contingencies will be assessed.

The model is currently in the testing phase. The first simulations on the developed code have been run and are currently being reviewed and adapted. It is expected that the first results will be shared with TSOs in early 2026.

In parallel to this work, Workstream 3B has been investigating other types of impact assessment already implemented by some other TSOs.

Impact assessment by Swissgrid

Swissgrid has developed an internal POC tool named ROSPRA (Robust Outage Scheduling using Probabilistic Risk Assessment) to support awareness and validation of planned outages across multiple planning horizons, ranging from year-ahead to week-ahead. The tool addresses significant uncertainties arising from system contingencies and nodal consumption and generation variations at these planning horizons. ROSPRA quantifies the contribution of individual planned outages to the overall system risk, computed based on the impacts of contingencies and the likelihood of associated nodal scenarios. This information enables outage planners to proactively revise their decisions well in advance, rather than deferring risks to real-time operations. In addition, this tool enables the identification of potential hidden risks associated with exceptional contingencies through targeted simulation.

ROSPRA comprises several key components designed to support probabilistic risk-based outage planning. For long-term planning, weather effects are excluded. Instead, a Markov chain-based model estimates generic failure rates per kilometre of transmission line, which are geographically adjusted across Switzerland using historical disturbance data.

The tool uses historical data to model vertical load and generation uncertainties. Techniques include copula families, Gaussian mixture models and Latin hypercube sampling. The impacts of contingencies and nodal scenarios are calculated using a Monte Carlo power-flow-based cascading simulation tool. A robust power flow simulation tool is required to analyse system behaviour under stress and to avoid unnecessary power flow divergence. Additionally, the simulation accounts for islanding scenarios and the distribution of imbalances, particularly in the event of system splits or unserved energy. Cascading simulations are considered by both fast and slow cascade mechanisms. Fast cascades are triggered by protection systems such as overcurrent protection or special protection schemes, while slow cascades involve prolonged element overloading, the availability of remedial actions and the time required for their activation. Finally, the prototype translates the complex probabilistic environment into a deterministic decision-making process by employing a traffic light-based situational awareness representation.

German TSO exceptional contingencies impact analysis

In the German transmission grid, exceptional contingencies (ECs) are defined annually based on their impact on both the national and neighbouring grid. The underlying methodology aims to assess previously identified common-cause failure scenarios by German TSOs to determine their relevance as ECs. The analysis focuses on identifying failure scenarios with supra-regional significance that must be considered when the likelihood of multiple failures increases. Due to their low probability, ECs are generally excluded from routine operational planning. However, they are included in grid calculations when there are indications of temporarily elevated failure risk.

The analysis considers the potential for cascading failures across control areas through iterative failure simulations. In each iteration, overloaded lines and transformers – defined as exceeding 100 % of their rated capacity – are assumed to fail, and the process continues until no further overloads occur or one of the following EC criteria is met:

- Cross-border impact: The initial failure or resulting disconnections cause overloads beyond specified limits in at least one foreign control area.
- Significant power loss: Disconnections due to failure or overload result in a cumulative loss of feed-in or load of at least 2,000 MW.
- System instability: The failure or subsequent disconnections lead to non-convergence in load flow calculations, indicating voltage instability or a critically stressed grid.

The total lost feed-in or load is calculated by summing the power of all units disconnected due to the initial failure or cascading outages, including voltage loss at the affected busbar. To ensure robust analysis, historical hourly snapshots spanning one year are used, reflecting varying switching states and transmission tasks. Each predefined failure scenario is simulated across all snapshots.

The results include identified ECs per snapshot, specifying the initial failure scenario, its control area and the amount of lost feed-in or load. Additionally, all lines disconnected during the iterative process are listed along with their utilisation levels at the time of disconnection.

3.3 Progress of WG PRA on data collection

3.3.1 PRA data collection

Since the second biennial PRA report, the working group has updated the *grid disturbance definitions for the power system above 100 kV* to refine the definitions and clarity of the contents. The next step is to update the grid disturbance profile with the recent updates in collaboration with ENTSO-E's Common Information Model Working Group (CIM WG). The grid disturbance profile is compatible with the CIM and allows detailed and harmonised reporting of grid disturbances.

Compatibility with the CIM ensures that PRA can be incorporated into existing operational processes in the future.

The 2025 survey showed that five more TSOs have improved their processes and are now fully ready to consistently collect information on grid disturbances in line with the ENTSO-E grid disturbance definitions and the grid disturbance profile.

3.3.2 Data collection and quality assurance

In 2025, 27 out of 40 TSOs provided grid disturbance data for 2023. Nevertheless, nine TSOs still provided only a partial dataset and might need support to provide the full set of information on grid disturbances, and two TSOs asked for dedicated assistance to complete the full data collection. WG PRA will continue to support TSOs in data collection and host dedicated webinars for them. Some TSOs have been unable to deliver grid disturbance data due to resource constraints, technical issues or insufficient data collection.

The working group has assessed the grid disturbance data received for its quality and consistency. Most improvements observed to date are related to identifying the causes of faults. However, "unknown" fault causes comprise around 17 % of all reported faults. In these cases, the motivations behind using the "unknown" fault cause are: 1) it is not worth allocating manual effort to investigate every successful auto-reclosing if automatic cause identification fails, and 2) some TSOs use "unknown" in case it is not 100 % certain. WG PRA continues to emphasise the importance of reporting the cause with the combined certainty level of the cause in cases when the certainty level is not 100 %.

Furthermore, a focus on the 220 kV and higher grid has been set because fault coverage in the 100 – 150 kV power grid is sparse, as many TSOs have no ownership or supervision over the assets in that voltage range. Additionally, the 100 – 150 kV grid is often interlaced with the distribution system operator (DSO) grids, making data collection difficult.

Given that the data collection process is not automated at all TSOs, it might take longer for some. Nevertheless, manual data validation is always needed to ensure high-quality data is used for the final processes. TSOs will need time and resources to adapt their collection processes to ensure high-quality data is used in the PRA methodology. WG PRA already has plans to direct additional support towards training material and to reduce manual processing times to guarantee the best results in terms of data consistency and quality of the collected grid disturbance data. WG PRA will continue to work on these points and provide targeted support to TSOs that are behind in the progress. In 2025, one webinar was organised for all TSOs. In the future, webinars or workshops will be held at least once a year and WG PRA will do deeper analysis on the received data to further improve its quality.

4 Ongoing PRA Activities Outside WG PRA

WG PRA also aims to be the centre of all PRA activities among TSOs. Indeed, some TSOs have initiated PRA activities outside of WG PRA. The group hosted a workshop in January 2025 to invite TSOs to present their activities and identify synergies among the various projects.

The methodologies and projects were assessed based on applicability, data requirements, benefits and limitations, with a focus on their relevance to WG PRA. Indeed, although within the scope of a PRA, some activities do not align with the WG PRA's legal requirements, while others could contribute to developing the PRA methodology. In the upcoming year, WG PRA will continue to engage with PRA activities to remain informed of progress and initiate further collaboration where relevant. This chapter briefly explains the ongoing PRA activities.

VAFFEL concept for failure-probability computations by Statnett [5]

Developed by Statnett and tested by MAVIR and TenneT NL, VAFFEL is a probabilistic tool for forecasting weather-related failures in transmission lines. The tool provides real-time risk assessments utilising Bayesian updating, historical failure data and medium-range weather forecasts.

While it has proven effective in modelling wind and lightning failures, with MAVIR extending its use to include icing, its reliance on high-quality weather data and its limitation to probability calculations without impact assessment are noted challenges. TSOs with significant wind-related failures could benefit from adopting VAFFEL, provided high-quality data is available.

PRA methodology by ELES

The ELES concept supports long-term grid planning by assessing the likelihood that system alterations will impact grid stability. While it provides useful insights for investment decisions, its reliance on historical datasets without real-time weather forecasting makes it less suitable for WG PRA applications.

ELES tested the concept, although its simplicity and lack of detailed probabilistic modelling limit its broader applicability. It remains useful for long-term planning but is not recommended for real-time operational risk assessment.

PRA methodology for outage planning by RTE

The RTE method is used for long-term security analysis in outage planning. The method performs N-1 analysis across various load and generation scenarios, considering long-term outage planning, nuclear group outages and net transfer capacity with neighbouring countries to identify challenging N-1 situations.

The approach can be summarised in three steps:

- Weather scenario simulations with the Antares-Simulator, accounting for maintenance planning, net transfer capacity and probabilistic weather scenarios. The simulator uses these inputs to generate 200 yearly adequacy scenarios of load, generation, unit availability and exchanges.
- Load flow simulation using IMAGRID to create simulation cases at an hourly timestep for each of the 200 yearly scenarios. N-1 simulations are performed on all cases with monitoring of the constraints.
- Results analysis to identify problematic N-1 cases and highlight the adequacy situations leading to such cases.

The results allow the planners to validate or adapt the long-term outage planning. The project has been implemented this year.

The PRA project at Swissgrid

The work initiated at Swissgrid focuses on outage planning, from yearly to weekly planning. The project is in the concept phase and aims to model uncertainties to identify and mitigate the risks associated with outage planning. The uncertainties considered include the nodal load or generation (modelled using historical data analysis), unexpected outages (modelled using a Markov chain and historical data), generator availability, line ratings (modelled using historical data and clustering of the grid based on weather conditions) and forced outages. The objective is to minimise the risk of an outage planning under such uncertainties using non-sequential Monte Carlo simulation.

PRA project by German TSOs

The project began in May 2024, and it aims to develop a tool for robust decision-making procedures based on PRA. The uncertainties considered refer to load and generation uncertainties, while the failure probability of grid components is out of scope. The project aims to determine operational risks, taking into account those uncertainties, and to develop the optimal set of measures and actions to be enforced when necessary.

PRA methodology for long-term planning by PSE

PSE uses a long-term planning tool to identify the critical network elements that must be protected, made redundant or prepared for fast replacement.

This tool performs a resilience assessment using a cascade simulation engine that accounts for the uncertainty of planned and unplanned outages, demand, wind and solar generation, import and export flows, etc. The tool groups data into clusters representing the evaluation horizon to manage uncertainties. Evaluating these uncertainties is computationally intensive, but the tool supports parallelisation on high-performance computing systems.

The cascade simulations are based on the DC model for load flow calculation and incorporate the Polish system defence plan, which provides the operator with guidance during a cascade event. In addition, the cascade simulation includes the relevant SPS that might react within the cascade, as well as the automatic and manual actions to be performed by the operator. The engine automatically identifies the most relevant failures of the system and their combinations.

The cascade simulation monitors the volume of redispatch and ENS at the end of the cascade, among other metrics. Later, it applies the failure probability to obtain a risk measure that defines the critical network elements in the grid.

For short-term impact assessment, this methodology can be adapted by removing the uncertainty handling module, since, for short-term forecasts can reasonably assume known conditions for outages and injections. In this case, only the cascade simulation engine is used.

Conclusion

At present, only the VAFFEL concept has been tested, although WG PRA will continue to engage with those TSOs to monitor their projects as they offer valuable insights into risk assessment, impact evaluation and contingency modelling. Further exploration of these projects could enhance the development of probabilistic risk assessments within ENTSO-E.

5 Future Roadmap

WG PRA has set the groundwork for developing the PRA methodology and will dedicate the following months to further developing and finalising the POC.

The POC will be tested on scenarios of the pan-European power system. The results will help determine whether and when the PRA methodology provides a better evaluation of contingencies than the N-1 criterion. Indeed, the theoretical concepts of the PRA methodology need to be assessed under realistic power system conditions to determine their added value. Comparing the contingency lists as determined today with the PRA risk screening list will enable a comparison of the two methods. The results will be presented to the TSOs – especially the operators – to clarify how they can use the methodology.

However, some open questions remain, including:

- When and how will PRA be implemented? As mentioned earlier, the method's reliability depends on the quality of the data, which should be considered when setting an implementation deadline. Additionally, the results will allow comparing the PRA with the N-1 criterion to decide on the implementation strategy: if both methods lead to similar results, a yearly re-assessment of the PRA could be envisioned, as well as defining rules and processes to determine when the PRA has an added value compared to the "N-1" criterion. The benefits of the PRA methodology might depend on the system's characteristics and topology. A parallel run of PRA with N-1 would help TSOs assess the risk level at which they are currently operating their grid. All these considerations need to be accounted for to develop an implementation strategy that is realistic and grounded in relevant results.
- How will PRA be used, and how will the operators and planners of the power system make use of the results? It is necessary to strike a balance between relevant indicators to assess the power system state, enabling fast and efficient decision-making.
- How can the impact of a contingency be reduced? How can the cost be accounted for in the risk assessment? Given that remedial actions to reduce the impact of a contingency often come with costs, different kinds of risk indicators related to the cost can be included in the scope, including the risk of a contingency without any actions, the risk with remedial actions and the risk with preventive measures. Optimal grid operation simultaneously minimises costs and risks. Therefore, the TSOs could also decide to adjust the acceptable risk level based on the cost of securing the grid against the contingency.

WG PRA next steps

WG PRA will continue to investigate failure-probability computations and test them when possible. Based on these investigations, TSOs will be able to choose the best-suited option for implementing failure-probability computations.

Additionally, WG PRA will continue to assist the TSOs with data collection. The new and improved grid disturbance definitions for the 100 kV and higher power grid will be published. WG PRA will periodically review data quality and host webinars or workshops for the TSOs to ensure that all necessary data for the probability or impact assessment is collected. As those will be used as inputs to the PRA methodology, all TSOs need to be onboarded on the process.

The topic of incorporating PRA in security assessment is a complex matter. Various PRA methodologies can be developed, whereby the probabilities can represent multiple variables, such as the probability of failure, uncertainties in load and generation, wind and sun forecasts, etc.

At the WG PRA level, it has been decided to focus on methodologies that account for the probability of failure of components of the grid as mandated in SO Regulation Article 71(1)(b) and 75(5). Those probabilities will be based on weather forecasts. Many of the TSOs' initiatives presented in Chapter 4 focus on uncertainties in generation and load, which lie beyond the scope of WG PRA.

At this stage, PRA methodology development remains in the domain of research and development (R&D). It holds strong importance that TSOs are aware of all PRA initiatives across the TSO community to ensure knowledge sharing, identify synergies between projects, investigate the possibility of merging methodologies, etc. For this purpose, WG PRA continues to engage with each TSO on their PRA initiatives and will plan workshops to share experience and results.

Finally, one of the key areas in the next stages is external stakeholder management to ensure the development of the PRA methodology aligns with regulators' expectations.

A first workshop with ACER and national regulatory authorities (NRAs) was held in mid-2025, and additional workshops will be held to follow up on progress. There is increasing interest in PRA among the TSOs, although not all views align with the legal requirements.

Additionally, it is important that regulators understand TSOs' starting points regarding PRA and their individual issues to plan a PRA methodology that could be applied at a pan-European level.

The high-level timeline in Figure 4 below illustrates WG PRA activities. Due to the complexity of planning, the timeline is subject to change as the work evolves. The CSAM methodology will be updated in line with progress in developing the PRA methodology.

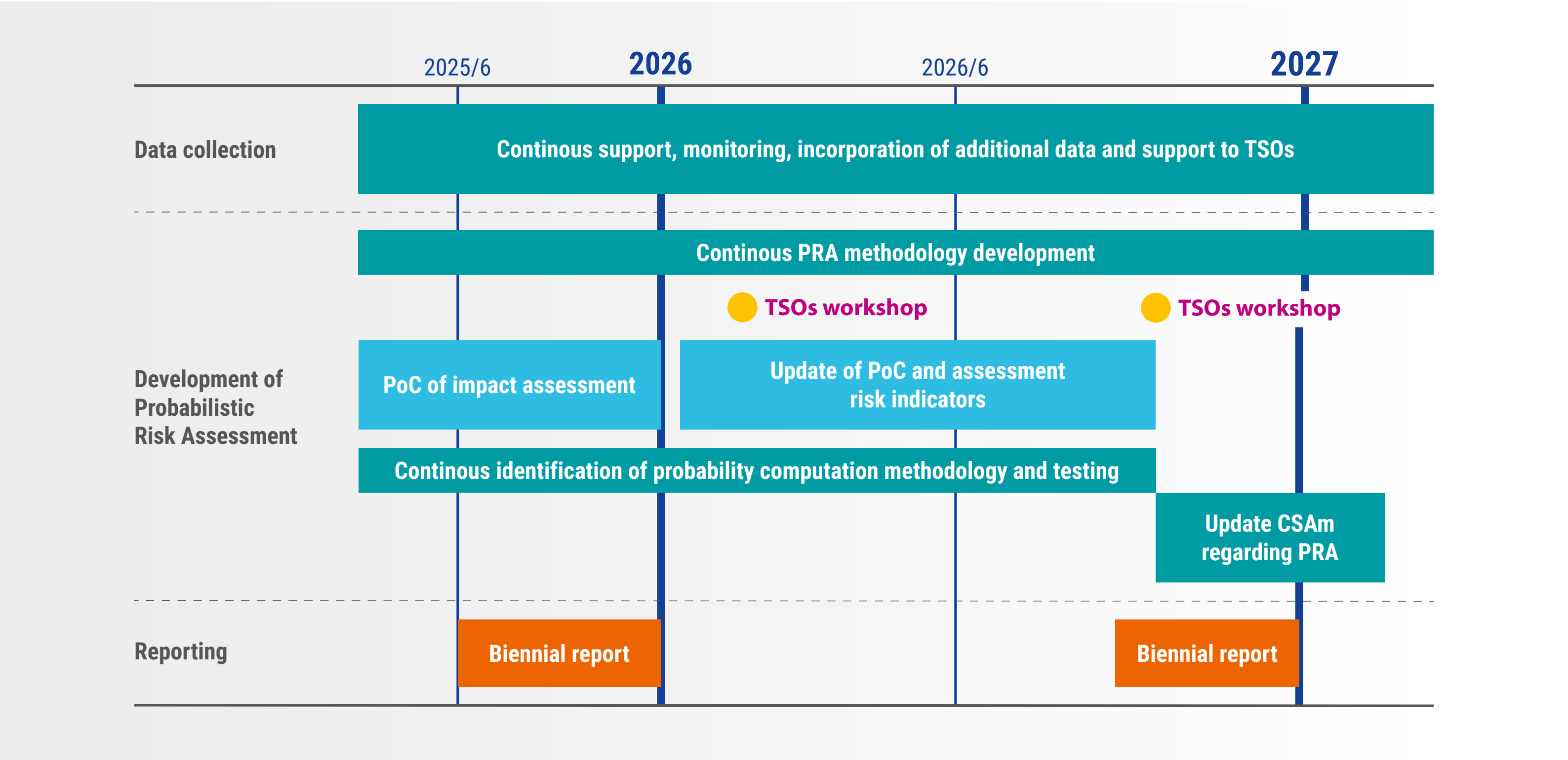


Figure 4: High-level timeline of the development process of the data collection, PRA methodology and administrative tasks (subject to change depending on the evolution of the work).

Conclusion

Consistent with the legal mandate in CSAM Article 44, ENTSO-E has supported all TSOs in investigating and preparing for a move towards a probabilistic risk assessment approach for the power grid, as a potential complement to the currently used 'N-1' criterion.

Current advancements include:

- launching the impact assessment's POC;
- thorough testing of the VAFFEL algorithm and evaluating other available probability calculation methods;
- continuation and improvement of the data collection process on the ENTREC platform, and support for TSOs;
- investigating all ongoing PRA activities in the TSO community.

In the upcoming months, WG PRA will focus on the POC development and testing on various power system states. The results will be analysed and discussed with all TSOs to assess the added value of the impact assessment developed in the scope of the PRA methodology. The next milestone for WG PRA will be to couple this impact assessment with reliable failure-probability computations to produce a robust risk assessment.

As previously mentioned, these probability computations are complex because they rely on large, reliable datasets and mathematical developments that must capture the specificity of various failure causes.

It is expected that the PRA methodology will be drafted by 2027 as a collaborative effort by all TSOs, with implementation occurring post-2027. WG PRA will continue to engage with the regulators at each relevant milestone to ensure transparency and alignment.

WG PRA would like to invite relevant parties (e.g., research institutes or other bodies) to get in touch if they wish to discuss their PRA-related research. To contact WG PRA, please email us at PRA@entsoe.eu.

Abbreviations

ACER	Agency for the Cooperation of Energy Regulators
CIM	Common information model
CSAM	Coordinating operational security analysis methodology
DSO	Distribution system operator
EC	Exceptional contingency
ENS	Energy not supplied
ENTSO-E	European Network of Transmission System Operators for Electricity
NRA	National regulatory authority
POC	Proof-of-concept
PRA	Probabilistic risk assessment
R&D	Research and development
RCC	Regional coordination centre
ROSPRA	Robust outage scheduling using probabilistic risk assessment
SPS	Special protection scheme
TSO	Transmission system operator
VoLL	Value of lost load
VAFFEL	Varsel før feil (NO), warning before failure (EN)
WG PRA	Working Group Probabilistic Risk Assessment

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[5] VAFFEL concept for failure-probability computations by Statnett

Publisher

ENTSO-E AISBL
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Design

DreiDreizehn GmbH, Berlin
www.313.de

Images

Cover: stock.adobe.com/peterschreiber.media
p. 5: iStock.com/DKosig
p. 13: iStock.com/Maximusnd

Publishing date

December 2025

European Network of
Transmission System Operators
for Electricity

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